

EPIDURAL ANESTHESIA (EA)

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Introduction to Epidural Anesthesia

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- EA produces a reversible loss of sensation and motor function much like a spinal with the exception that LA is placed within the epidural space.
- Larger doses of local anesthetic are required to produce anesthesia when compared to a spinal anesthetic.
- Doses must be monitored to avoid toxicity.

Introduction to EA...

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- An epidural catheter allows the versatility to extend the duration of anesthesia beyond the original dose by the administration of additional LA.
- EA can be combined with GA or used as the sole anesthetic agent
- EA can be segmental
 - The block is most intense near the site of catheter or needle insertion and diminishes with distance

EA Indications

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- ❑ Labor Analgesia
- ❑ Cesarean section
- ❑ Procedures of the uterus, perineum
- ❑ Hernia repairs
- ❑ Genitourinary procedures
- ❑ Lower extremity orthopedic procedures
- ❑ For those who may not tolerate a general anesthetic
- ❑ Peri-operative analgesia for major thoraco-abdominal surgeries,...

- Should NOT be used in patients who are hypovolemic or severely dehydrated.
- Patients should be pre-hydrated with 0.5 – 1 liter of crystalloid solutions immediately prior to the block.

EA Advantages

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- Easy to perform (though it takes a bit more experience than SA)
- Provides excellent operating conditions
- The ability to administer additional local anesthetics increases duration
- The ability to use the epidural catheter for postoperative analgesia

Advantages...

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- Return of gastrointestinal function generally occurs faster than with GA
- Patent airway
- Fewer pulmonary complications compared to general anesthesia
- Decreased incidence of DVT and pulmonary emboli formation compared to general anesthesia

EA - Disadvantages

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- Risk of block failure
 - The rate of failure is slightly higher than with a SA
 - Always be prepared to induce general anesthesia if block failure occurs.
- Onset is slower than with spinal anesthesia.
 - May not be a good technique if there is little time to properly perform the procedure.

Disadvantages...

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- Alteration in the patient's BP and HR
 - generally slower onset with less alteration in blood pressure and heart rate than with a spinal anesthetic
 - It is essential to place the epidural block with monitoring of at least, blood pressure and pulse oximetry.
 - Resuscitation medications/equipment should be available.
- There is a decreased complication rate compared to SA though.

Disadvantages...

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- Continuous epidural catheters should be used with caution on the ward if the patient's vital signs are NOT closely monitored.
- Risk for infection, resulting in serious complications.

Absolute Contraindications

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- ☐ Patient refusal
- ☐ Infection at the site of injection
- ☐ Coagulopathy
- ☐ Severe hypovolemia
- ☐ Increased Intracranial pressure
- ☐ Severe Aortic and/or mitral Stenosis...

Relative Contraindications

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- Sepsis
- Uncooperative patients
- Pre-existing neurological deficits
- Demyelinating lesions
- Moderate stenotic valvular heart lesions
- Severe spinal deformities

Controversial

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- Prior back surgery
- Inability to communicate with the patient
- Complicated surgeries that may involve:
 - prolonged periods of time to perform,
 - major blood loss,
 - maneuvers that may complicate respiration

Mechanism/Site of Action

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- Administered at a physiologic distance when compared to spinal anesthesia
- The intended targets are the spinal nerves and associated nerve roots
- Several barriers to the spread of local anesthetic to the intended site of action results in the requirement of larger volumes of local anesthetic when compared to spinal anesthesia

Mechanism/Site of Action...

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- A segmental block is possible because the anesthetic can be confined close to the level at which it was injected.
- A segmental block is characterized by a well-defined band of anesthesia at certain nerve roots; leaving nerve roots above and below unblocked.
- This can be seen with a thoracic epidural that provides upper abdominal anesthesia while sparing cervical and lumbar nerve roots.
- Usually it is dose dependent

Barriers

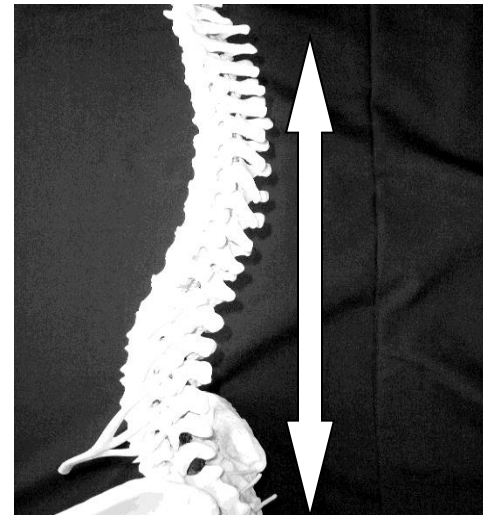
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- Dura mater between the epidural space and spinal nerve and nerve roots act as a modest barrier
- The majority of the solutions is absorbed systemically through the venous rich epidural space
- Epidural fatty tissue acts as a reservoir which can prolong the duration
- The remainder reaches the spinal nerve and nerve roots.

Spread of LA in the Epidural Space

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- Local anesthetic injected into the epidural space moves in a horizontal and longitudinal manner.
- Theoretically the longitudinal spread could reach the foramen magnum and sacral foramina if enough volume was injected.



Spread of Local Anesthetics- Horizontal

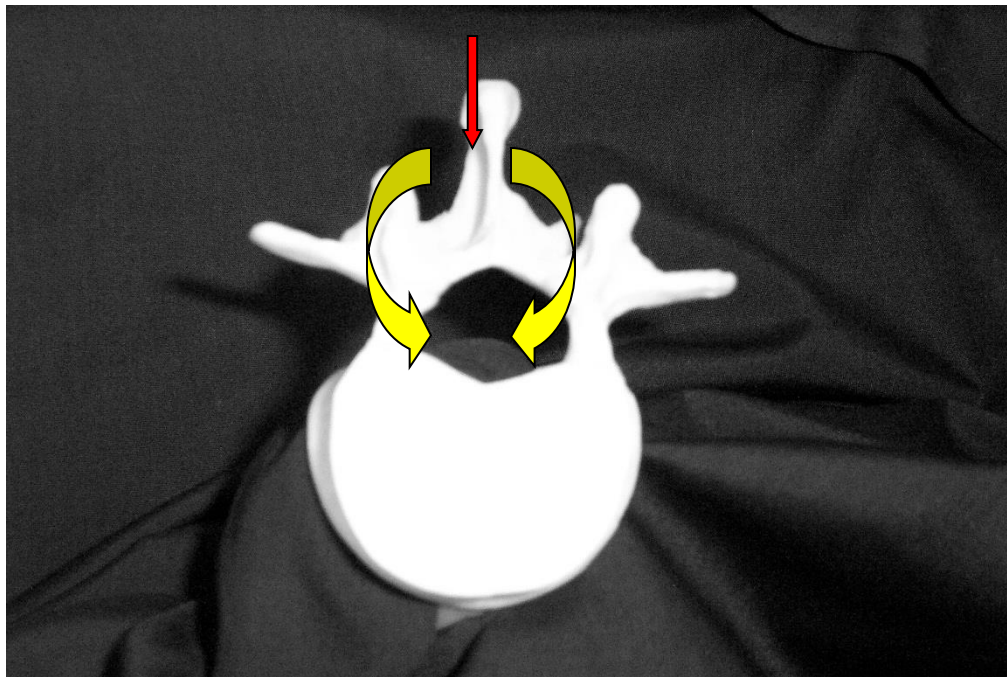
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- Horizontally the LA spreads through the intervertebral foramina to the dural cuff
- LAs spread through the dural cuff via the arachnoid villa and into the CSF
- Blockade occurs at the mixed spinal nerves, dorsal root ganglia, and to a small extent the spinal cord.

Spread of LA – Horizontal...

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- The horizontal spread of LAs gives access to nerve roots exiting laterally



Factors affecting spread of LA

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- ☐ Injection site
- ☐ Volume
- ☐ Concentration
- ☐ Position
- ☐ Age

Preparation

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- An epidural kit includes essential equipment for administering Epidural anesthesia:
 - Epidural needle
 - Epidural catheter
 - Catheter Connector
 - Filter

Epidural Needles



Hustead



Tuohy



Crawford

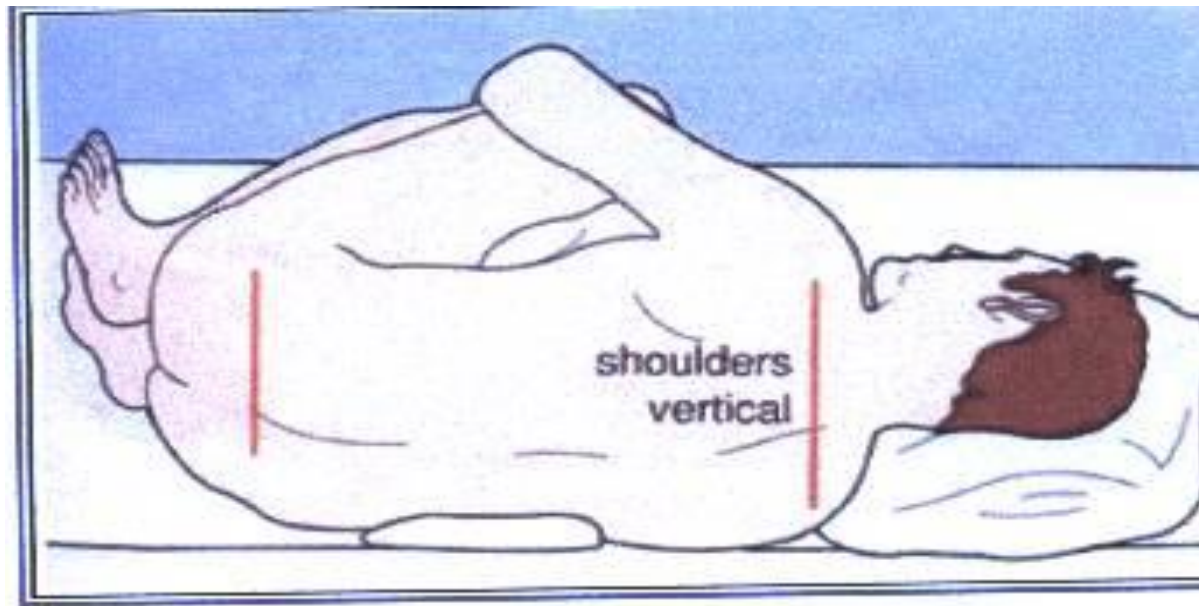
Combined Spinal/Epidural Needle



Position

Patient assume a sitting or side-lying position with the back arched toward the provider.

Help to spread the vertebrae apart



Mid calf position

In the mid-calf position, the patient rests the lower legs (mid-calf), rather than the knees, on the edge of the bed, sitting somewhat further back on the bed than in the conventional sitting position.

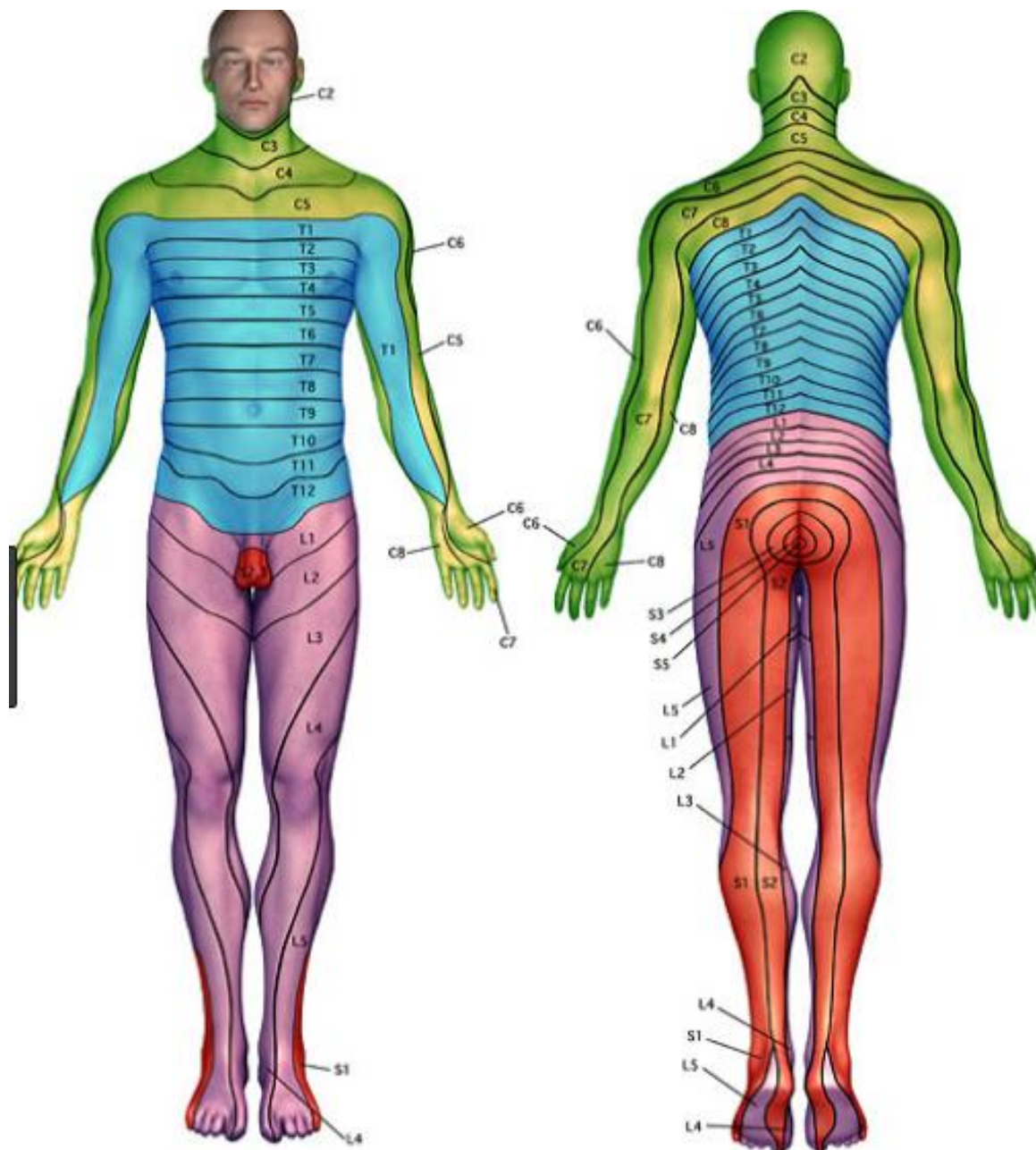
As a result, the knees are slightly flexed with the patient's back nearer to the practitioner.

The patient's neck is flexed forward and the arms are crossed in front of the body



- One advantage of the mid-calf position is that the patient naturally assumes an ideal position for placement of a neuraxial block with little instruction.
- The shoulders fall forward and the flexed position achieved appears to optimally open the spaces between the spinous processes

- The site is dependent upon the area of pain
- Epidural catheter should be placed in the middle of the dermatomes affected by surgery
- Incision Target dermatome level >> Insertion
- Thoracic T4-T6 >>>> T8-T10 vertebral level
- Upper abdo T6-T8 >>> T10-T12 Vert level
- Lower abdo T8-T10 >>> T12-L2 level
- Pelvic T10-T12 >>> L2 – L5 vert level
- Lower extremity L1-L4 >>> L2-L5



MAP KEY

L1,2,3,4	Anterior and inner surface of lower limbs
L4,5, S1	Foot
L4	Medial side of great toe
S1,2,L5	Posterior and outer surface of lower limbs
S1	Lateral margin of foot and little toe
S2,3,4	Perineum
T10	Level of umbilicus
T12	Inguinal or groin regions
C5	Clavicles
C5, 6, 7	Lateral parts of upper limbs
C8, T1	Medial sides of the upper limbs
C6	Thumb
C6, 7, 8	Hand
C8	Ring and little fingers
T4	Level of nipples

Projection and Puncture

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- When a lumbar approach is used, the depth from skin to the Ligamentum flavum (LF) commonly approaches 4 cm, with the depth in most patients being between 3.5 and 6 cm.
- The Epidural space is 5 to 7 mm thick in the midline, which requires needle control if unintentional dural puncture is to be prevented.
- Some investigators are exploring the use of ultrasound to help minimize dural puncture by more accurately defining the distance between the skin and the epidural space.

Projection and Puncture...

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- When a thoracic approach is chosen, control is of greater importance because injury to the spinal cord is possible if the needle is advanced too far.
- Clinically, thoracic epidural anesthetics do not appear to be associated with an increased incidence of neurologic injury because those choosing to use the technique are most often anesthesiologists with considerable experience in lumbar epidural anesthesia.

- ❑ Ideal Placement (adult) 7-10 cm at the skin
- ❑ Epidural catheters have markings that indicate their length.
 - ❑ there is a mark at the tip of the catheter
 - ❑ the 1st single mark up the catheter is 5cm
 - ❑ double mark up the catheter is 10 cm
 - ❑ triple mark on the catheter is 15 cm
 - ❑ four mark together indicate 20cm
- ❑ A change in depth of the catheter indicates migration either into or out of the epidural space.



Technical Difficulty

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- ❑ **The fatness** - most problematic: get a harpoon and a lucky charm
- ❑ **Old people** - calcified ligaments: you may have to abandon procedure
- ❑ **Prior back surgery** - Heavy Metal
- ❑ **Autoimmune + collagen d/o** - have ligaments like paper don't slip or you might get a spinal tap



Transforaminal Epidural Depth by Body Mass Index

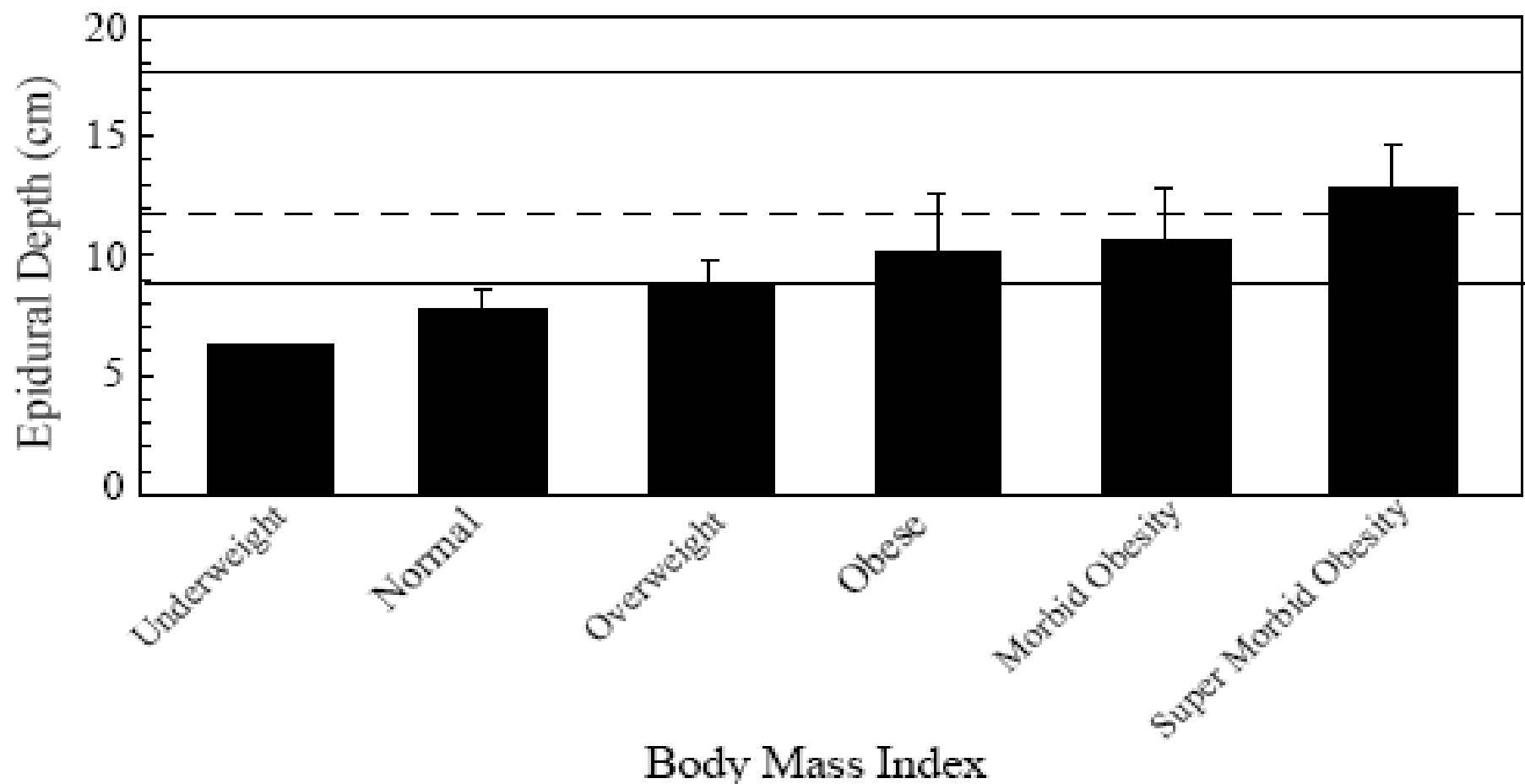


Figure A. The depth of needle lumbar transforaminal epidural needle insertion is associated with body mass index (BMI). Obese patients (BMI 30-34.99) had a significantly increased depth when compared with normal patients (BMI < 25). Morbidly obese patients (BMI 35-39.99) had a significantly increased depth when compared to normal and overweight patients. Super-morbid obesity (BMI > 40) was associated with an increased depth when compared with normal, overweight and obese patients.

- = Length of 3.5-inch needle (8.89 cm)
- — — = Length of 5-inch needle (12.7 cm)
- — — = Length of 7-inch needle (17.78 cm)

Distribution, Uptake & Elimination

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- Takes 6-8 times the dose of a spinal anesthetic to create a comparable block.
 - **Larger mixed nerves** are found in the epidural space when compared to the subarachnoid space.
 - Local anesthetics are lipid soluble and will be absorbed by tissue and **epidural fat**.
 - **Epidural veins** absorb a significant amount of local anesthetic with blood concentrations peaking in 10-30 minutes after a bolus.

Distribution, Uptake & Elimination...

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- Local anesthetics absorbed in the epidural veins will be diluted in the blood.
- The pulmonary systems acts as a temporary buffer and protects other organs from the toxic effects of LA to some extent.
- Distribution occurs to the vessel rich organs, muscle, and fat.

Distribution, Uptake & Elimination...

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- Long acting amides will bind to alpha-1 globulins which have a high affinity to local anesthetics but become rapidly saturated.
- Amides are metabolized in the liver and excreted by the kidneys.
- Esters are metabolized by pseudocholinesterase so rapidly that there are rarely significant plasma levels.

Factors Affecting Height of Epidural Blockade

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- Volume of local anesthetic
- Age
- Height of the patient
- Gravity

Volume

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- Can be variable
- General rule: 1-2 ml of local anesthetic per dermatome
- i.e. epidural placed at L4-L5; you want a T4 block for a C-sec. You have 4 lumbar dermatomes and 8 thoracic dermatomes. $12 \text{ dermatomes} \times 1-2 \text{ ml} = 12-24 \text{ ml}$
- Big range! Stresses importance of incremental dosing!

Volume...

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- If you require only segmental anesthesia, then the dose would be less.
- Volume of local anesthetic plays a critical role in block height.
- Dose of LAs administered in thoracic area should be decreased by 30-50% due to decrease in compliance and volume.

Age

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- As age increases the amount of local anesthetic to achieve the same level of anesthesia decreases. A 20 year old vs 80 year old
- This is due to changes in size and compliance of the epidural space

Height

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- The shorter the patient the less local anesthetic required.
- A patient that is only 5'3" (160cm) may require 1 ml per dermatome while someone who is 6' (180cm) may require the full 2 ml per dermatome

Gravity

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- Position of patient does affect spread and height of local anesthetic BUT not to the point of spinal anesthesia.
- i.e. lateral decubitus position will “concentrate” more local anesthetic to the dependent side will a weaker block will occur in the non-dependent area.
- A sitting patient will have more local anesthetic delivered to the lower lumbar and sacral dermatomes

Gravity

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- L5-S2 sometimes will have ‘patchy’ anesthesia due to sparing. By having the patient on “sitting” position, one can concentrate local anesthetic to this area.
- Trendelenberg or reverse trendelenberg may help spread local anesthetic cephalad or alternatively limit the spread.

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